

5-run local-exec provisioner to print the public_ip of bastion ec2

```
aws_instance.app: Creation complete after 17s [id=i-0fd440df1a52bc5d0]
aws_instance.bastion: Provisioning with 'local-exec'...
aws_instance.bastion (local-exec): Executing: ["/bin/sh" "-c" "echo bastion IP
]
aws_instance.bastion (local-exec): bastion IP is 3.237.100.182
aws_instance.bastion: Creation complete after 18s [id=i-0c37d5c053b9cb5bd]
```

6- upload infrastructure code on github project

7-verify your email in ses service

```
resource "aws_ses_email_identity" "email" {
  email = "janna.wael.ali@gmail.com"
}
```

AWS Business Applications > Amazon Simple Email Service

Congratulations!

You have successfully verified an email address. You can now start sending email from this address.

For new Amazon SES users

If you have not yet applied for an Amazon SES account, please see [the setup page](#).

8- create lambda function to send email

Functions (1)

Last fetched 5/3/2026, 5:21:35 PM

Actions

Create function

Q Search by attributes or search by keyword

<1>

☐	Function name	Description	Package type	Runtime	Type	Last modified
☐	terraform-state-notifier	-	Zip	Python 3.11	Standard	1 minute ago

9-create trigger to detect changes in state file and send the email

to me

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

Report not spam

Hello,

Your Terraform state file has been modified.

Bucket : state-terr-bucket

File : env%3A/prod/terraform.tfstate

This means terraform apply was just run
and your infrastructure has changed.

Regards,
AWS Notification

10-create rds (mysql)

myrdsinstance Modify Actions

Summary

DB Identifier myrdsinstance	Status Available	Role Instance	Engine MySQL Community	Recommendations
CPU 7.02%	Class db.t3.micro	Current activity 0 Connections	Region & AZ us-east-1a	

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I accessed it from bastion ec2

```
[ec2-user@ip-10-0-5-67 ~]$ mysql -h myrdsinstance.cy368qy8kj0n.us-east-1.rds.amazonaws.com -P 3306 -u myrdsuser -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 5.7.44-rds.20260212 MySQL 5.7 is now under RDS Extended Support. Please upgrade to 8.0 to benefit from the latest RDS innovations.

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```

11-create elastic cache (redis)

my-redis-cluster Info Upgrade to Valkyrie Modify Actions

Cluster details

Cluster name my-redis-cluster	Description -	Node type cache.t3.micro	Status Available
Engine Redis	Engine version 7.0.7	Global datastore -	Global datastore role -
Update status Up to date	Cluster mode Disabled	Shards 0	Number of nodes 1
Data tiering Disabled	Multi-AZ Disabled	Auto-failover Disabled	Encryption in transit Disabled
Encryption at rest Disabled	Parameter group default.redis7	Outpost ARN -	Configuration endpoint -
Primary endpoint -	Reader endpoint -	ARN arn:aws:elasticache:us-east-1:531472034848:cluster:my-redis-cluster	Data migration No active migrations

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I accessed it from bastion ec2 i tried Set a key-value pair and then retrieve it also i tested the connection

```
[ec2-user@ip-10-0-5-67 ~]$ redis6-cli -h my-redis-cluster.jbavty.0001.usel.cache.amazonaws.com -p 6379
-bash: redis6-cli: command not found
[ec2-user@ip-10-0-5-67 ~]$ redis6-cli -h my-redis-cluster.jbavty.0001.usel.cache.amazonaws.com -p 6379
-bash: redis6-cli: command not found
[ec2-user@ip-10-0-5-67 ~]$ redis-cli -h my-redis-cluster.jbavty.0001.usel.cache.amazonaws.com -p 6379
my-redis-cluster.jbavty.0001.usel.cache.amazonaws.com:6379> ping
PONG
my-redis-cluster.jbavty.0001.usel.cache.amazonaws.com:6379> SET mykey "Hello, ElastiCache!"
OK
my-redis-cluster.jbavty.0001.usel.cache.amazonaws.com:6379> GET mykey
"Hello, ElastiCache!"
my-redis-cluster.jbavty.0001.usel.cache.amazonaws.com:6379> █
```